SOLUTION DESIGN DOCUMENT

Invoice Processing Automation

Accounts Payable · UiPath REFramework · Document Understanding · SQL

Document Information	
Field	**Details**
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Version	1.0 — Initial Release
Author	Blessing Nnabugwu, RPA Developer
Date	March 2026
Tool / Platform	UiPath Studio, UiPath Orchestrator, SQL Server
Architecture	REFramework with Orchestrator Queue Management
Status	DRAFT — For Review

Version History			
Version	**Date**	**Author**	**Description**
1.0	March 2026	Blessing Nnabugwu	Initial draft

1. Executive Summary

This Solution Design Document (SDD) describes the design, architecture, exception handling strategy, data mapping, and deployment approach for the Invoice Processing Automation bot developed using UiPath Studio and deployed via UiPath Orchestrator.

The automation targets the Accounts Payable function, where staff currently process incoming vendor invoices manually — extracting data from PDF documents, validating against purchase orders in the ERP system, and updating the financial database. This process is high-volume, repetitive, and error-prone.

The bot uses UiPath Document Understanding to extract structured data from PDF invoices, validates each record against a SQL database, routes exceptions for human review, and writes a full audit log on every transaction. It is built on the UiPath REFramework pattern with Orchestrator queue management for scalability and resilience.

Business Impact Summary	
Metric	**Value**
Manual effort eliminated	~95% reduction in processing time per invoice
Processing speed	Under 2 seconds per invoice (vs. 4–6 minutes manually)
Error rate	Near-zero on clean documents (vs. ~3% human error rate)
Audit coverage	100% — every transaction logged with full context
Volume capacity	500+ invoices per scheduled run

2. Process Overview

2.1 AS-IS Process (Current State)

The current invoice processing workflow is performed entirely by the Accounts Payable team. The process is manual, time-consuming, and subject to human error, particularly during peak periods when invoice volumes are high.

Current Process Steps

1. AP team receives vendor invoices via email or shared drive as PDF attachments.
2. Staff member opens each PDF and manually reads invoice number, vendor name, date, line items, and total amount.
3. Staff queries the ERP system to locate the corresponding Purchase Order (PO).
4. Manual comparison of invoice values against PO values is performed.
5. If values match, invoice is approved and payment record is updated in the SQL database.
6. If values do not match, invoice is escalated to a supervisor for review.
7. All actions are logged manually in a spreadsheet — subject to omissions.

AS-IS Pain Points

Pain Point	Business Impact
High manual effort — 4–6 min per invoice	Staff time diverted from higher-value tasks
Human data entry errors (~3% error rate)	Payment delays, supplier relationship issues
No real-time audit trail	Compliance risk, difficult to investigate disputes
Process bottleneck during peak periods	Invoice backlogs, late payments, penalty charges
Inconsistent escalation handling	Exceptions handled differently by different staff

2.2 TO-BE Process (Automated State)

The automation replaces all manual steps with a UiPath bot operating on the REFramework pattern. The bot initialises from configuration, retrieves invoice items from an Orchestrator queue, processes each transaction, and handles exceptions systematically. Human involvement is limited to reviewing flagged exceptions and approving escalated cases.

TO-BE Process Steps

8. Bot initialises — loads config from Orchestrator Assets and Excel config file. Validates application availability.
9. Queue population — a separate dispatcher bot (or scheduled job) scans the invoice inbox and adds each PDF path as a queue item in Orchestrator.
10. Bot retrieves queue item — fetches next pending transaction from the Orchestrator queue.
11. Document Understanding — extracts structured data from the PDF invoice using UiPath Document Understanding and ABBYY OCR engine.
12. SQL validation — queries the database to locate the matching Purchase Order and validates invoice fields against PO data.

13. Decision: if validation passes — updates payment record in SQL, writes success log, marks queue item as Successful.
14. Decision: if business exception — marks queue item as Failed with reason code, writes to exception log, sends email alert to AP supervisor.
15. Decision: if application exception — retries up to configured maximum attempts. If retry limit exceeded, marks queue item as Failed, sends alert, and bot continues to next item.
16. End of queue — bot sends summary report email and terminates cleanly.

3. Scope

3.1 In Scope

- Reading PDF invoice files from a designated shared folder or SharePoint location
- Extraction of invoice fields using UiPath Document Understanding
- SQL database queries to retrieve and validate Purchase Order records
- Updating payment status records in the SQL database on successful validation
- Business exception handling: missing PO, amount mismatch, duplicate invoice
- Application exception handling: system unavailability, timeout, login failure
- Full transaction audit log written to SQL on every queue item
- Email notification to AP supervisor on business exceptions and end-of-run summary
- Orchestrator queue management, scheduling, and bot monitoring
- UAT support and post-deployment hypercare (2 weeks)

3.2 Out of Scope

- Automated payment execution — bot validates and records, but does not trigger bank payments
- Processing of invoices received in formats other than PDF (e.g., paper, Word, XML)
- Integration with external vendor portals
- Invoice approval workflow beyond routing exceptions to supervisor email
- Changes to the ERP system or SQL schema

4. Solution Architecture

4.1 Framework

The bot is built on the UiPath Robotic Enterprise Framework (REFramework) — the industry-standard architecture for transactional automation. REFramework manages initialisation, transaction processing, exception handling, and clean shutdown through a structured state machine.

REFramework States

State	Description
Initialisation	Load config (Orchestrator Assets + Excel). Open applications. Validate system availability. Retry up to 3 times before aborting.
Get Transaction	Retrieve next queue item from Orchestrator. If no items remain, transition to End Process.
Process Transaction	Execute core business logic: extract, validate, update, log. Handle business and application exceptions separately.
End Process	Close all applications, send summary email, log end-of-run statistics to Orchestrator.

4.2 Components

Component	Type	Purpose
Main.xaml	REFramework state machine	Orchestrates the full process flow
InitAllApplications.xaml	Workflow	Opens UiPath, SQL connection, validates environment
GetTransactionData.xaml	Workflow	Fetches next item from Orchestrator queue
Process.xaml	Workflow	Core logic — extract, validate, update, log
ExtractInvoiceData.xaml	Reusable workflow	Document Understanding extraction pipeline
ValidateAgainstPO.xaml	Reusable workflow	SQL query and field-level comparison logic
WriteAuditLog.xaml	Reusable workflow	Inserts audit record to SQL on every transaction
SendExceptionAlert.xaml	Reusable workflow	Emails AP supervisor with exception detail
CloseAllApplications.xaml	Workflow	Graceful shutdown of all open applications
Config.xlsx	Config file	Environment settings, thresholds, email addresses

5. Data Mapping

5.1 Invoice Fields Extracted

The following fields are extracted from each PDF invoice using UiPath Document Understanding. All extracted values are validated for presence and format before processing continues.

Field Name	Source	Format	Mandatory	Validation
Invoice Number	PDF — header	String (alphanumeric)	Yes	Not null, unique
Vendor Name	PDF — header	String	Yes	Match against vendor master
Invoice Date	PDF — header	Date (DD/MM/YYYY)	Yes	Valid date, not future
PO Number	PDF — line items	String	Yes	Exists in PO table
Line Item Description	PDF — line items	String	No	Informational only
Line Item Amount	PDF — line items	Decimal (2dp)	Yes	Greater than 0
Tax Amount	PDF — totals	Decimal (2dp)	No	Non-negative
Total Amount	PDF — totals	Decimal (2dp)	Yes	= Sum of line items + tax

5.2 SQL Tables Referenced

Table	Operation	Purpose
dbo.PurchaseOrders	SELECT	Retrieve PO details for validation against invoice fields
dbo.Vendors	SELECT	Validate vendor name and ID against vendor master table
dbo.InvoicePayments	SELECT / INSERT / UPDATE	Check for duplicate invoices; insert approved payment record
dbo.RPA_AuditLog	INSERT	Write audit record on every transaction — success, failure, or exception
dbo.RPA_ExceptionLog	INSERT	Write detailed exception record for all failed transactions

6. Exception Handling Design

All exceptions are classified into one of two categories — Business Exceptions and Application Exceptions — each handled differently within the REFramework.

6.1 Business Exceptions

Business exceptions occur when the data itself is invalid or does not meet processing rules. They are not caused by system failures. The bot does not retry business exceptions — it logs the reason, marks the queue item as Failed, and alerts the AP supervisor.

Exception Type	Trigger Condition	Bot Action
Missing PO Number	PO Number field is null or blank after extraction	Log + fail queue item + email alert
PO Not Found	PO Number extracted but no matching record in dbo.PurchaseOrders	Log + fail queue item + email alert
Amount Mismatch	Invoice total differs from PO approved amount by more than tolerance threshold	Log + fail queue item + email alert
Duplicate Invoice	Invoice Number already exists in dbo.InvoicePayments	Log + fail queue item + email alert
Invalid Date	Invoice date is in the future or unparseable format	Log + fail queue item + email alert
Vendor Mismatch	Vendor name on invoice does not match vendor master record for that PO	Log + fail queue item + email alert
Extraction Confidence Low	Document Understanding confidence score below 80% threshold	Route to human review queue

6.2 Application Exceptions

Application exceptions are caused by system or environment failures — not data issues. The bot retries these up to the configured maximum before escalating.

Exception Type	Trigger Condition	Bot Action
Login Failure	Cannot log into application within timeout period	Retry init up to 3× then abort
SQL Connection Error	Database connection string fails or times out	Retry transaction up to config max
Application Timeout	UI element not found within configured wait time	Retry transaction up to config max
File Not Found	PDF path in queue item does not exist or is inaccessible	Fail queue item, log path, alert
Extraction Engine Error	Document Understanding pipeline throws unhandled error	Retry once, then fail and alert

Retry configuration is managed in Config.xlsx. Default: maximum 3 retries per transaction. After exhausting retries, the bot marks the queue item as Failed with a detailed error message and stack trace, then moves to the next item.

7. Security & Compliance

- All database credentials are stored as Orchestrator Assets — never hardcoded in any workflow file.
- Bot runs under a dedicated service account with least-privilege access to SQL tables (read on PO/Vendor tables; read-write on Invoice/Audit tables only).
- PDF invoice files are not modified or deleted by the bot — read-only file access.
- All audit log entries are write-once — no update or delete permissions on dbo.RPA_AuditLog.
- Exception emails contain invoice reference numbers only — no full invoice content transmitted by email.
- Bot execution is restricted to scheduled windows via Orchestrator — no ad hoc execution permitted without authorisation.
- Document Understanding models are stored in AI Center with version control — model changes require re-validation before production use.

8. UAT Test Cases

The following test cases must be executed and signed off by the business before the bot is approved for production deployment.

TC#	Test Scenario	Input	Expected Result	Pass/Fail
TC-01	Happy path — clean invoice	Valid PDF, matching PO, correct totals	Queue item Successful, payment record inserted, audit log written	
TC-02	Missing PO Number	PDF with PO field blank	Business exception raised, queue item Failed, supervisor email sent	
TC-03	PO not in database	PDF with valid-format PO not in dbo.PurchaseOrders	Business exception — PO Not Found, queue item Failed	
TC-04	Amount mismatch	Invoice total exceeds PO approved amount by >tolerance	Business exception — Amount Mismatch, queue item Failed	
TC-05	Duplicate invoice	Invoice Number already in dbo.InvoicePayments	Business exception — Duplicate, queue item Failed, no duplicate insert	
TC-06	Low extraction confidence	Scanned/low-quality PDF	Routed to human review queue, not processed	
TC-07	SQL connection failure	SQL server unavailable (simulated)	Application exception, retry up to max, then Failed with stack trace	
TC-08	Empty queue	No items in Orchestrator queue	Bot initialises, finds no items, sends summary email, closes cleanly	
TC-09	Multiple invoices — bulk run	Queue with 20 mixed items (clean + exceptions)	Each item processed independently, correct outcome per type	
TC-10	Audit log completeness	Run TC-09 batch	Exactly 20 audit log records written — one per queue item	

8.1 UAT Sign-Off

Role	Name	Signature / Date
RPA Developer	Blessing Nnabugwu	
AP Team Lead / Business Owner		
QA Lead		
IT / Infrastructure		

9. Deployment & Go-Live Plan

9.1 Pre-Deployment Checklist

- All UAT test cases passed and signed off
- Production Orchestrator queue created and tested
- Orchestrator Assets configured with production credentials
- Bot service account created with correct SQL permissions
- Shared invoice folder/SharePoint path accessible from bot machine
- Email notification distribution list confirmed with AP team lead
- Config.xlsx updated with production values and uploaded to Orchestrator
- Rollback plan documented and agreed with IT

9.2 Schedule

Phase	Activity	Owner
Week 1	UAT execution and defect resolution	Blessing Nnabugwu + AP Team
Week 2	UAT sign-off and pre-deployment checklist	All stakeholders
Week 3	Production deployment — limited run (50 invoices)	Blessing Nnabugwu + IT
Week 4–5	Hypercare — monitor all runs, daily check-in with AP team	Blessing Nnabugwu
Week 6	Hypercare exit — handover to support team with runbook	Blessing Nnabugwu + Support

10. Post-Deployment Monitoring

After go-live, bot health is monitored via UiPath Orchestrator dashboards and the SQL audit log. The following KPIs are tracked weekly during hypercare and monthly thereafter.

KPI	Target	Alert Threshold
Queue item success rate	≥ 97%	Alert if below 95% in any single run
Business exception rate	< 5%	Investigate if above 8% — may signal data quality issue
Application exception rate	< 1%	Immediate review if above 2% — system stability concern
Average processing time per invoice	< 2 seconds	Alert if average exceeds 5 seconds
Audit log completeness	100%	Zero tolerance — every queue item must have an audit record

10.1 Runbook Summary

A separate Operational Runbook is provided for the support team covering: how to restart a failed run, how to requeue failed items, how to update Orchestrator Assets, known issues and resolutions, and escalation contacts.

— End of Document —